

Practical No : 1

Aim: To implement Linear Regression

```
✓ [11] import pandas as pd
0s      import numpy as np
      import matplotlib.pyplot as plt
      from sklearn.linear_model import LinearRegression
      from sklearn.metrics import mean_absolute_error, mean_squared_error
```

```
✓ [4] df = pd.read_csv('/content/Experience-Salary.csv')
```

```
[ ] df.head()
```



	exp(in months)	salary(in thousands)
0	18.290293	16.521825
1	17.023407	11.666234
2	26.343613	23.167255
3	19.105834	20.877145
4	27.742516	23.166236

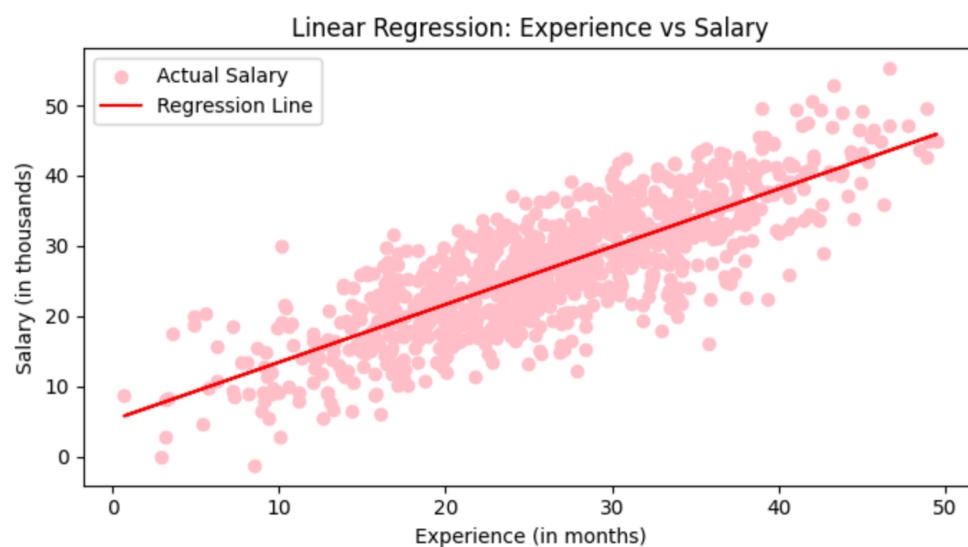
```
✓ [5] # create an object
0s      regression=LinearRegression()
```

```
✓ [6] X = df[['exp(in months)']]
0s      y = df[['salary(in thousands)']]
```

```
✓ [7] # Train the model
0s      model = LinearRegression()
      model.fit(X, y)
```

```
✓ # Predict
0s df['Predicted salary (in thousands)'] = model.predict(X)
```

```
✓ # Plot
0s plt.figure(figsize=(7, 4))
    plt.scatter(df[['exp(in months)']], y, color='pink', label='Actual Salary')
    plt.plot(df[['exp(in months)']], df[['Predicted salary (in thousands)']], color='red', label='Regression Line')
    plt.xlabel('Experience (in months)')
    plt.ylabel('Salary (in thousands)')
    plt.title('Linear Regression: Experience vs Salary')
    plt.legend()
    plt.tight_layout()
    plt.show()
```



0s



#Evaluation

```
print(f'Mean Absolute Error : {mean_absolute_error(y, df["Predicted salary (in thousands)"])}')  
print(f'Mean Squared Error : {mean_squared_error(y, df["Predicted salary (in thousands)"])}')  
print(f'Root Mean Squared Error : {np.sqrt(mean_squared_error(y, df["Predicted salary (in thousands)"])}')
```



```
Mean Absolute Error : 4.087007093719527  
Mean Squared Error : 26.75353064566882  
Root Mean Squared Error : 5.172381525532394
```